

NEXUS

Relationship Layer — System Atlas

NEXUS is a lightweight semantic lens for BIM models.

It does **not** replace Revit, ACC, BIMcollab, Speckle, or any CDE. Instead, it exposes relationships that already exist — or *should* exist — but are currently invisible, fragmented, or implicit.

The **Relationship Layer** is the first layer of NEXUS. It focuses on answering a simple but powerful question:

“What is related to what — and *why*?”

1. Why NEXUS Exists

Modern BIM tools are excellent at geometry and parameters, but weak at expressing **meaningful relationships**:

- A face recognition device can be placed near a door, but cannot *declare* that it controls that door
- A system exists conceptually, but is scattered across families, parameters, and views
- Responsibility (設計 / 生産設計 / 施工 / 保守) is known socially, but not structurally
- Vendor and subcontractor boundaries are discussed in meetings, not models

NEXUS exists to: - Make these relationships explicit - Keep them lightweight and optional - Avoid introducing yet another workflow-heavy system

2. What NEXUS Is (and Is Not)

NEXUS IS

- A semantic visualization layer
- A relationship graph derived from BIM data
- A tool you open *when you need clarity*
- Friendly to Japanese and international workflows

NEXUS IS NOT

- A CDE
 - A model authoring tool
 - A rule engine
 - A replacement for existing platforms
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3. Core Concept: Relationship Layer

The Relationship Layer expresses **typed connections** between nodes.

Nodes represent: - BIM Elements (doors, devices, equipment) - Spaces / Rooms - Systems - Organizations (設計者, 元請, サブコン) - Vendors / Manufacturers

Edges represent **relationships**, not geometry.

4. Relationship Types (Initial Set)

Relationship types are intentionally simple, readable, and extensible.

4.1 Physical / Spatial

- **HostedOn** - element is hosted by another
- **LocatedIn** - element exists within a space
- **AdjacentTo** - proximity without control

4.2 Functional / System

- **Controls** - logical control relationship
- **Supplies** - power, signal, or resource supply
- **ConnectedTo** - physical or logical connection

4.3 Responsibility / Process (Japan-aware)

- **DesignedBy** - 基本設計 / 実施設計
- **EngineeredBy** - 生産設計
- **BuiltBy** - 施工
- **MaintainedBy** - 保守

4.4 Commercial / Organizational

- **ProvidedBy** - vendor or manufacturer
- **InstalledBy** - subcontractor
- **OwnedBy** - asset ownership

4.5 Meta

- **DependsOn**
- **DerivedFrom**

Each relationship can optionally carry: - Direction - Confidence (Confirmed / Inferred) - Notes

5. System Atlas (Primary View)

System Atlas is the default visualization of the Relationship Layer.

It is not a diagram. It is a **map**.

Characteristics: - Systems act as visual anchors - Elements orbit systems - Organizations and vendors sit at the boundary

System Atlas answers: - “What makes up this system?” - “What does this device affect?” - “Who is involved, directly or indirectly?”

6. Layout Modes (Thinking Modes)

NEXUS offers multiple layouts, each representing a way of thinking.

Default (Organic)

- Free exploration
- Discovery-focused

System Atlas

- System-centric understanding
- Best for MEP / security / controls

Responsibility Atlas

- Organization-centric
- Excellent for Japanese project structures

Spatial Stack

- Room / space centric
- Useful for FM and operations

Layouts do not change data — only perspective.

7. Navigation & UX Principles

NEXUS follows three rules:

1. You should never get lost
2. You should never be forced to commit
3. You should always understand what you are seeing

Key features: - Mini map overview - Fit / Center / Reset controls - Zoom limits - Focus-on-select behavior - Light / Dark themes

8. JSON Schema Philosophy

NEXUS uses a simple, explicit JSON structure.

Design goals: - Human-readable - Diff-friendly - Exportable from Revit or other tools - No required IDs from external systems

The schema favors clarity over compression.

9. Typical Use Cases

- System clarification during coordination
 - Explaining scope boundaries in meetings
 - Checking responsibility gaps
 - Onboarding new team members
 - Design → construction → operation handover
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9.1 Concrete JSON Example

Below is a **minimal but realistic JSON example** that demonstrates how the Relationship Layer works in practice.

This example models: - An entrance door - A face recognition device - An access control system - A room
- Design, construction, and vendor responsibility

```
{
  "nodes": [
    {
      "id": "door_entrance_01",
      "label": { "en": "Entrance Door", "jp": "エントランス扉" },
      "category": "Element",
      "subCategory": "Door",
      "system": "Access Control"
    },
    {
      "id": "device_face_recognition_01",
      "label": { "en": "Face Recognition", "jp": "顔認証装置" },
      "category": "Element",
      "subCategory": "SecurityDevice",
      "system": "Access Control"
    },
    {
      "id": "system_access_control",
      "label": { "en": "Access Control System", "jp": "入退室管理システム" },
      "category": "System"
    }
  ],
}
```

```

{
  "id": "room_lobby",
  "label": { "en": "Lobby", "jp": "ロビー" },
  "category": "Room"
},
{
  "id": "org_architect",
  "label": { "en": "Architectural Team", "jp": "建築設計" },
  "category": "Organization",
  "role": "Design"
},
{
  "id": "org_subcon_security",
  "label": { "en": "Security Subcontractor", "jp": "セキュリティサブコン" },
  "category": "Organization",
  "role": "Construction"
},
{
  "id": "vendor_abc_security",
  "label": { "en": "ABC Security Co.", "jp": "ABCセキュリティ" },
  "category": "Vendor"
}
],
"edges": [
  {
    "source": "device_face_recognition_01",
    "target": "door_entrance_01",
    "type": "Controls",
    "directional": true,
    "confidence": "Confirmed"
  },
  {
    "source": "device_face_recognition_01",
    "target": "room_lobby",
    "type": "LocatedIn",
    "directional": false
  },
  {
    "source": "door_entrance_01",
    "target": "system_access_control",
    "type": "PartOf",
    "directional": false
  },
  {
    "source": "device_face_recognition_01",
    "target": "system_access_control",
    "type": "PartOf",
    "directional": false
  },
  {

```

```

    "source": "door_entrance_01",
    "target": "org_architect",
    "type": "DesignedBy",
    "directional": false
  },
  {
    "source": "device_face_recognition_01",
    "target": "org_subcon_security",
    "type": "BuiltBy",
    "directional": false
  },
  {
    "source": "device_face_recognition_01",
    "target": "vendor_abc_security",
    "type": "ProvidedBy",
    "directional": false
  }
]
}

```

How to Read This Example

- Nodes are **entities**, not geometry
- Systems are explicit, not inferred
- Responsibility is expressed as relationships, not parameters
- The same structure works across design, construction, and operation

This JSON can be: - Exported from Revit - Manually authored - Generated incrementally

10. Roadmap (Intentional and Realistic)

Phase 1 – Relationship Layer (Current)

- Graph visualization
- Typed relationships
- Multiple layouts
- Themes and navigation

Phase 2 – Authoring Assist (Optional)

- Manual relationship editing
- Notes and annotations
- Relationship confidence tagging

Phase 3 – Model Integration (Selective)

- Revit add-in sync
- Round-trip element selection
- Parameter hints (not enforcement)

Phase 4 – Additional Layers (Conceptual)

- **Intent Layer** – why something exists
- **Lifecycle Layer** – design → build → operate
- **Risk Layer** – single points of failure

Each layer is optional and independent.

11. Design Ethos

NEXUS follows these principles:

- Small surface area
 - Respect existing workflows
 - Visual clarity over automation
 - No forced adoption
 - Useful even if used alone
-

12. Final Note

NEXUS is deliberately modest.

Its strength is not in doing everything, but in revealing what is already there — and helping people talk about it clearly.

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